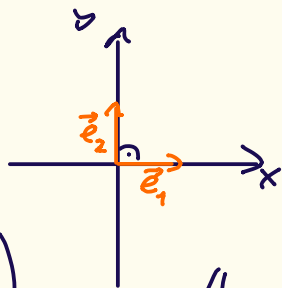


2.2

$$B_1 = \left(\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right)$$



$$R_\varphi = \begin{pmatrix} \cos \varphi & \sin \varphi \\ -\sin \varphi & \cos \varphi \end{pmatrix}$$

báze po rotaci o φ úhel

$$B_2 = \left(\underset{\parallel \vec{v}_1}{\begin{pmatrix} \cos \varphi \\ \sin \varphi \end{pmatrix}}, \underset{\parallel \vec{v}_2}{\begin{pmatrix} -\sin \varphi \\ \cos \varphi \end{pmatrix}} \right)$$

$$A = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$$

Jelikož rotace zachovává délky i úhly, pak

B_2 je stále ON

$$0 = \langle \vec{v}_1, \vec{v}_2 \rangle = -a \cos \varphi \sin \varphi - b \sin^2 \varphi + b \cos^2 \varphi + c \cdot \cos \varphi \sin \varphi$$

$$0 = -a \frac{\sin 2\varphi}{2} + b \cdot \cos 2\varphi + c \cdot \frac{\sin 2\varphi}{2}$$

$$0 = (c - a) \frac{\sin 2\varphi}{2} + b \cos 2\varphi$$

$$k \in \mathbb{Z} \\ \varphi \neq \frac{\pi}{4} + k \frac{\pi}{2} \\ a \neq c \wedge \cos 2\varphi \neq 0$$

$$(a - c) \frac{\sin 2\varphi}{2} = b \cos 2\varphi \quad / : ((a - c) \cos 2\varphi) \frac{1}{2}$$

$$\frac{\sin 2\varphi}{\cos 2\varphi} = \frac{2b}{a - c}$$

$$\tan 2\varphi = \frac{2b}{a - c}$$

$$\varphi = \frac{1}{2} \arctan\left(\frac{2b}{a - c}\right) *$$

$$a = c$$

v

$$\cos 2\varphi = 0$$

$$\varphi = \frac{\pi}{4} + k \frac{\pi}{2}$$

$$\sin\left(\frac{\pi}{2} + k\pi\right) = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$0 = b \cos 2\varphi$$

$$\downarrow \quad \downarrow \\ b = 0 \vee \cos 2\varphi = 0$$

$$2\varphi = \frac{\pi}{2} + k\pi, k \in \mathbb{Z}$$

$$\varphi = \frac{\pi}{4} + k \frac{\pi}{2}$$

to stejné

1) k sudé

$$(a - c) \cdot \frac{1}{2} = 0$$

$$\frac{a - c}{2} = 0 \\ a = c$$

$$B_2 = \left(\begin{pmatrix} \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{pmatrix}, \begin{pmatrix} -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{pmatrix} \right)$$

$$a=c \wedge b=0$$

$$A = \begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix} = a I_2 \Rightarrow \text{2 Du 1.2.}$$

$\Rightarrow$ s. sk. s. vynásobený $a \Rightarrow a > 0$

$$B_2 = \left(\begin{pmatrix} -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{pmatrix}, \begin{pmatrix} +\frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} \end{pmatrix} \right)$$

plyne 2 Du 1.2.

$$A = \begin{pmatrix} a & b \\ b & a \end{pmatrix}, a > 0$$

$a^2 > b^2$

* z tohoto vždy dostanu
úhel φ když dostanu $\langle \cdot, \cdot \rangle$
a tím pádem i matici $A = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$.
Tedy jsme schopni vždy takovou
hledanou bázi najít. $\square$

2) kličo'

$$(a-c) - \frac{1}{2} = 0$$

$$a=c$$

$$B_1 = \left(\begin{pmatrix} -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{pmatrix}, \begin{pmatrix} -\frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} \end{pmatrix} \right)$$

$$B_2 = \left(\begin{pmatrix} +\frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} \end{pmatrix}, \begin{pmatrix} \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{pmatrix} \right)$$

$$A = \begin{pmatrix} a & b \\ b & a \end{pmatrix}, a > 0$$

$a^2 > b^2$